

1. Observe the acceleration of the tangent of the angle between the ground and your line of sight with the ball.
2. If the acceleration is positive, step backward.
3. If the acceleration is negative, step forward.
4. Repeat steps 1–3 until the ball is directly in front of your face.
5. Catch it.

One serious objection to Chapman's five-step method is that outfielders following this process seem to have to calculate the tangents of angles on the fly, meaning we're replacing an inner physicist theory with an "inner geometer theory" in which baseball players can instantaneously, and subconsciously, take tangents.

One potential resolution to this objection is that for many angles,  $\tan(\theta)$  is approximately equal to  $\theta$ , so rather than observing the acceleration of a tangent, outfielders can merely observe the acceleration of an angle. If the acceleration of an angle can be estimated by the felt acceleration of the neck joints that crick as the neck moves back to observe the ball, and if an angle is a reasonable approximation for its tangent, then we don't need to assume any great subconscious mathematical or geometrical powers on the part of outfielders—only the physical skill of being accurately attuned to subtle sensory inputs.

By making an acceleration estimate the only difficult part of the process, we have obtained a potential solution to the outfielder problem that has much more psychological plausibility than the inner physicist's theory of subconsciously extrapolated parabolas. Of course, the psychological appeal of the solution doesn't mean that it can be used only by humans. A robot outfielder could also be programmed to follow Chapman's five-step process, and it might even perform better at catching the ball if it did so, because, for example, Chapman's process enables those who use it to dynamically respond to changes due to wind or bounces.

Besides psychological plausibility, there's one more crucial feature that the five-step process implied by Chapman's insight possesses: it doesn't rely on a solve-for- $x$  strategy or any explicit equation at all. Instead, it proposes successive iterations of easy observations and small, gradual steps